

SOLVERS IN MACHINE LEARNING (Ridge Focus)

1 What is a Solver?

Definition:

A **solver** is the optimization algorithm used to find the model parameters (weights) that minimize the loss function.

In simple words:

Solver = Method used to compute optimal weights.

2 Why Do We Need a Solver?

In ML we minimize a loss function.

Example: **Ridge Regression**

$$\text{Loss} = ||y - Xw||^2 + \alpha ||w||^2$$

To minimize this loss, we must calculate **w**.

The algorithm used to compute this **w** = solver.

3 Types of Solvers (Big Classification)

There are 2 main types:

● A) Direct (Closed-form) Solvers

They use **linear algebra** to compute exact solution.

Example formula in Ridge:

$$w = (X^T X + \alpha I)^{-1} X^T y$$

Examples:

- `cholesky`
- `svd`

Characteristics:

- ✓ Exact solution
 - ✓ Fast for small datasets
 - ✗ Memory heavy
 - ✗ Not suitable for very large datasets
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● B) Iterative Solvers

They update weights step-by-step using optimization methods like Gradient Descent.

General update rule:

$$w = w - \eta \nabla L$$

Examples:

- `sag`
- `saga`
- `lsqr`
- `sparse_cg`

Characteristics:

- ✓ Good for large datasets
 - ✓ Memory efficient
 - ✓ Works well for sparse data
 - ✗ Approximate solution
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4 Important Ridge Solvers Explained

◆ 1. Cholesky Solver

Type:

Direct solver

How it works:

Uses **Cholesky Decomposition**:

If matrix A is positive definite:

$$A = LL^T A = LL^T A = LL^T$$

Instead of inverse, it solves system using triangular matrices.

When to use:

- Small to medium datasets
- When matrix is well-conditioned

Pros:

- ✓ Very fast
- ✓ Efficient for small problems

Cons:

✗ Can be unstable if matrix nearly singular

◆ 2. SVD (Singular Value Decomposition)

Type:

Direct solver

How it works:

Decomposes matrix into:

$$X = U \Sigma V^T \quad X^T = U^T \Sigma V = U \Sigma V^T$$

More numerically stable.

When to use:

- Multicollinearity
- Ill-conditioned matrix

Pros:

- ✓ Very stable
- ✓ Works even if matrix nearly singular

Cons:

✗ Slower than Cholesky

◆ 3. SAG (Stochastic Average Gradient)

Type:

Iterative solver

How it works:

Variant of gradient descent using averaged gradients.

When to use:

- Large datasets

Notes:

- Needs feature scaling
 - Faster convergence than basic GD
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◆ **4. SAGA**

Improved version of SAG.

- ✓ Supports L1 regularization
 - ✓ Faster convergence
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◆ **5. sparse_cg / lsqr**

Used for:

- Sparse matrices
- Very large datasets

Memory efficient.

5 Cholesky vs SVD (Interview Comparison)

Feature	Cholesky	SVD
Speed	Faster	Slower
Stability	Less stable	More stable
Multicollinearity	Not ideal	Good
Dataset size	Small	Small/medium

6 Direct vs Iterative (Important Interview Table)

Direct	Iterative
Exact solution	Approximate
Uses matrix decomposition	Uses gradient descent
Fast for small data	Scales to big data
Memory heavy	Memory efficient

7 When To Use Which Solver?

Small dataset → `cholesky`

Multicollinearity → `svd`

Huge dataset → `sag` / `saga`

Sparse data → `sparse_cg`

8 Interview-Ready Answer (Memorize This)

If asked:

What is a solver?

Say:

A solver is the optimization algorithm used to compute model parameters by minimizing the loss function. In Ridge regression, solvers can be direct methods like Cholesky and SVD that use matrix decomposition, or iterative methods like SAG and SAGA that use gradient-based optimization.

9 Advanced Understanding (For Strong Impression)

You can add:

- Ridge loss is convex → global minimum guaranteed.
 - Direct solvers compute closed-form solution.
 - Iterative solvers are better when feature count is very high.
 - Regularization term ensures matrix is positive definite → allows Cholesky.
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10 Very Important Concept Link

Since you built your own GDRegressor:

Your implementation = Iterative solver.

Sklearn Ridge with `cholesky` = Direct solver.

That's the core difference.